CHAPTER

8

Instruments, Instrumentation and Operating Position

With the evolution of tooth reduction methods, the cavity preparation is usually performed with the air turbine handpiece. The hand instruments account for only a small portion of the tooth reduction. Rotary burs or diamonds are used for gross reduction; whereas, hand-cutting instruments are used to produce intricate details of the cavity preparation. In the use of hand-cutting instruments, there is no vibration, pressure or pain element as compared to high-speed rotatory instruments.

Both hand and rotary-driven instruments are used in conservative dentistry. Hand instruments can further be cutting and non-cutting (general); whereas, rotary can be cutting (tooth preparation), abrasive (finishing), etc.

HAND INSTRUMENTS

Table 8.1 describes classification of hand instruments.

Table 8.1: Classification of hand instruments

A. ***Hand-cutting instruments***
a. Chisels
  i. Straight
  ii. Monoangle
  iii. Binangle
  iv. Triple angle
b. Excavators
  i. Hoe
  ii. Spoon
  iii. Cleoid-Discoid
c. Modified form of chisels
  i. Hatchet
  ii. Gingival margin trimmer
  iii. Angle former
  iv. Wedelstaedt chisel
d. Miscellaneous
  i. Knives
  ii. Dental files
B. ***Non-cutting (general) instrument***
a. Diagnostic instruments
  i. Mouth mirror
  ii. Probes
  iii. Tweezers
b. Restorative instruments
  i. Condensers
  ii. Carvers
  iii. Burnishers
  iv. Plastic instrument
c. Finishing/polishing instruments

A. HAND-CUTTING INSTRUMENTS

Hand-cutting instruments are mandatory for refining cavity preparation.

The standardization of hand-cutting instruments, is identified by a nomenclature and number; describing the size, design and common clinical usage.

The proficiency and quality of the clinical treatment depend on proper selection and use of the hand-cutting instruments.

Metallurgy

The hand-cutting instruments are generally manufactured from carbon steel/stainless steel.

Carbon steel: It is an alloy made by the addition of a small percentage of carbon to iron. There are two varieties of steel, viz. soft and hard.

Soft steel contains 0.5% or less carbon, while hard steel contains 0.5–1.5%, carbon. Other ingredients are manganese 0.2%, silicone 0.2% and the rest is iron (91.4–91.6%).

In order to improve upon physical characteristics, two heat treatments (hardening and tempering) are given to steel.

In hardening heat treatment, steel is heated approximately to 1500°F (815°C) in oxygen-free environment, then immediately plunged into cold water or oil. The hardening heat treatment hardens the alloy but makes it brittle.

In tempering heat treatment, steel is reheated for 1 hour at 350°F (176°C) and then plunged in cold water/slightly acidulated water/oil/mercury. Heat tempering relieves strain, decreases brittleness and increases toughness.

Various alloys of steel are used in manufacturing; the main are:

a. ***Stainless steel:*** It is composed of carbon 0.6–1.0%; chromium 18% and iron 81–81.4%. It is extremely hard; requires careful and precise heat treatment during manufacture. It loses a sharp edge easily as compared to carbon steel (used for handle, shank and part of blade).
b. ***Monel metal:*** A natural alloy of nickel, copper and iron, Monel metal contains nickel 67%, copper 28%, and iron 5%.
c. ***Nichrome:*** Nichrome contains nickel 60–80%, chromium 12–20% and iron 0–2.6%.
d. ***Stellite:*** Stellite consists of cobalt 65–90%, chromium 10–35%, with small quantities of tungsten, molybdenum, iron or nickel. Its chief characteristics are hardness, strength, density, high melting point, maintenance of cutting edge and resistance to action of acids.
e. ***Tarno:*** Tarno is made up of chromium alloy, used mainly for manipulation of the cements.

Design Characteristics

A hand-cutting instrument consists of the following parts:

a. ***Handle or shaft:*** It is mostly straight and octagonal in cross-section, and may be serrated to increase friction for hand gripping (Fig. 8.1).
Handles are available in various sizes and shapes. Mostly handles are in a continuation of the shank. If the shank and blade are separate from the handle and intended to be screwed into it, the instrument is known as a cone socket instrument. The advantage with the cone socket instrument is that, if the working end is broken, it may be easily replaced and the disadvantage is its tendency to loosen at the joint.

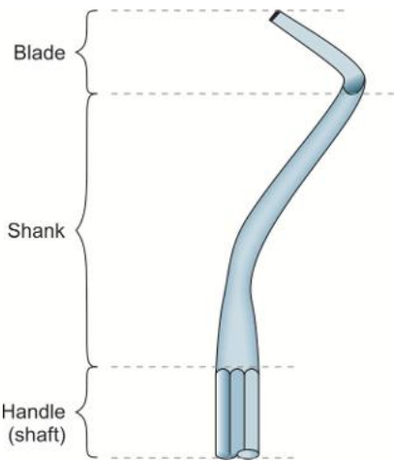


Fig. 8.1: Part of a hand instrument

b. ***Shank:*** It connects the handle with the blade (Fig. 8.1). It usually tapers from its connection with the handle. Any angulation in the instrument can be placed at the junction of shaft and shank.
c. ***Blade/nib:*** The working end of cutting instrument is called the blade; whereas, working end of the non-cutting instrument, such as condenser, is called a nib (Fig. 8.2A to F). The working surface of nib is called a face.
d. ***Cutting edge:*** The working part of the instrument, usually in the form of a bevel with different shapes (Figs 8.3 and 8.4).
e. ***Blade angle:*** The angle between the long axis of the blade and the long axis of the shaft (Fig. 8.3).
f. ***Cutting edge angle:*** The angle between the margins of the cutting edge and the long axis of the shaft (Fig. 8.3).

The angle of the shank is so designed that the cutting edge falls in the diameter of handle; feature allows concentration of force onto the blade without any rotation of the instrument: A few authors name it as 'Balancing'.

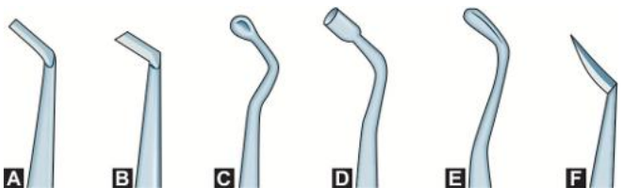


Fig. 8.2: Characteristic blades/nib of: **(A)** Hatchet; **(B)** Hoe; **(C)** Excavator; **(D)** Condenser; **(E)** Plastic instrument; **(F)** Carver

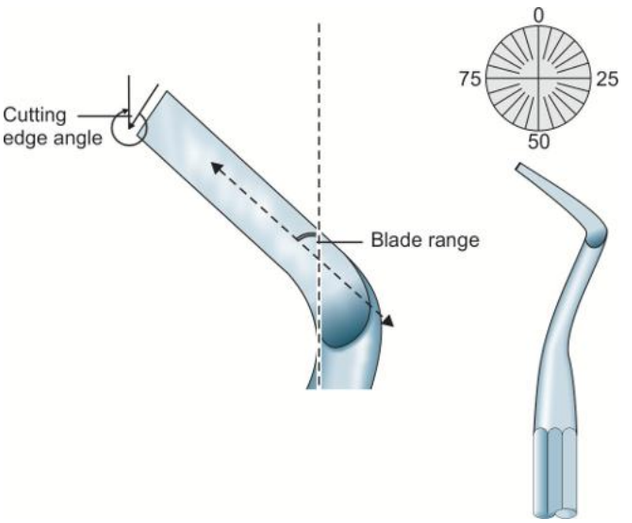


Fig. 8.3: Binangle hatchet showing blade angle, cutting edge and cutting edge angle (angle measured in centigrades; shown in circle)

Instrument Nomenclature

Dr GV Black established a nomenclature for hand instruments, similar to the biological classification.

1. *Order:* Purpose of instrument, e.g. excavator or scaler.
2. *Suborder:* Position or manner of use, e.g. push, pull.
3. *Class:* Form of working end, e.g. hatchet, chisel.
4. *Subclass:* Shape of the shank, e.g. monoangle, binangle.

Naming the instruments moves from 4 to 1, e.g. a binangle (subclass), hatchet (class), push (suborder), excavator (order). In most cases, the suborder describing the position or manner of use is variable and non-specific; may be omitted practically.

Instrument Formula

Dr GV Black proposed an instrument formula that describes dimension and angulation of the hand instruments (Fig. 8.4).

The instrument formula consists of three units, whose measurements are based upon the metric system:

i. The 1st figure represents the width of the blade in tenths of a millimeter.
ii. The 2nd figure represents the length of the blade in millimeters.
iii. The 3rd figure represents the angle, which the blade forms with the axis of the handle. This angle is expressed in 100th of a circle or centigrades.

These three measurements are sufficient for describing a great percentage of instruments. However, for instruments with their cutting edges at an angle other than a right angle to the long axis of the blade, a fourth unit, cutting edge angle, is added to the basic three-unit formula. This additional number represents the angle formed between the cutting edge and the long axis of the handle. It is placed in the second position of the formula, i.e. before the length of the blade. For example, a gingival margin trimmer has a 4 unit formula (Fig. 8.4).

Many a times, manufacturers also quote their number on the handle; this does not relate to the instrument formula.

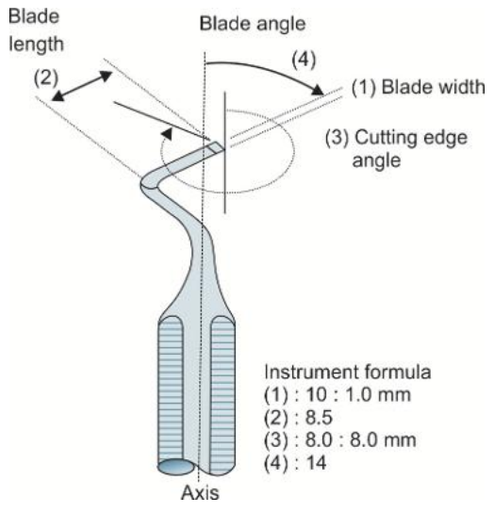


Fig. 8.4: Instrument formula

Cutting Plane (Instrument Beveling)

i. ***Single plane instruments:*** The instruments where force is applied in the same plane as that of the blade and handle is called 'single plane instrument'. Even if they may have two or more curves or angles in their shank, all are in the same plane as the handle (Fig. 8.5). They are mainly used for 'direct cutting'.

Direct cutting instruments are made 'left' or 'right' by placing a bevel on one side of the blade. If the instrument is held with the cutting edge down and pointing away from the operator, and bevel is on the right side, it will be a 'right-sided

Lateral cutting
• Curved plane (double plane) • Force is applied at a right angle to the plane of the blade and handle • The curves and the angle are in the plane at right angle to the handle • Double plane instruments are used only for lateral cutting
Direct cutting
• Straight blade (single plane) • Force is applied in the same plane as that of blade and handle • The curves and the angles in the shank are in the same plane as the handle • Single plane instruments can be used both for direct cutting and lateral cutting

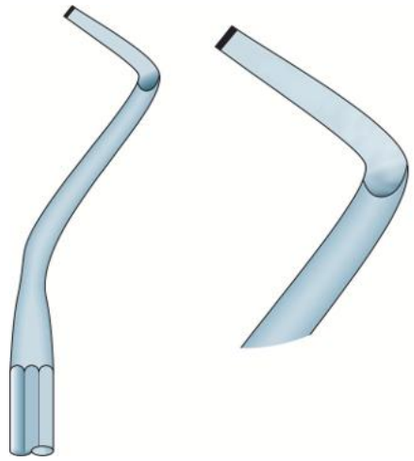


Fig. 8.5: Binangle hatchet with cutting edge in the plane of the instrument

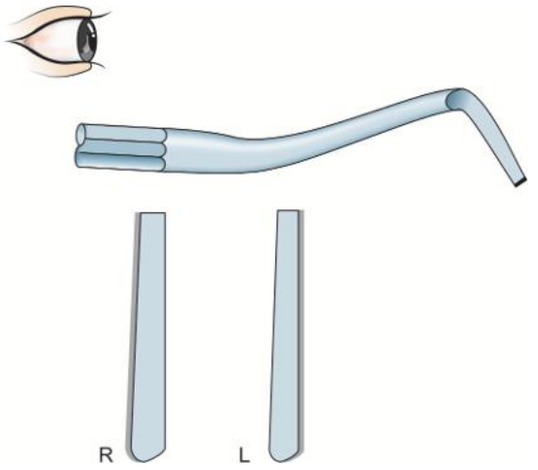


Fig. 8.6: To determine 'right' or 'left', hold the instrument with primary cutting edge pointing down and away; see the bevel

instrument'. If the bevel is on the left, it will be a 'left-sided instrument' (Fig. 8.6).

Single-plane instruments with no angle in the shank have the potential for five cutting movements, viz. vertical, right, left, push and pull.

A right-left single-plane instrument with one or more angles in the shank will have four potential cutting movements; viz. vertical, push, pull, right or left. For the mesially and distally beveled single-plane instruments, the four movements are vertical, right, left and push or pull, depending upon the location of the bevel.

ii. ***Double plane instruments:*** The instruments where the force is applied at a right angle to the plane of the blade and handle (usually have a curved blade) are called 'double plane instruments' (Fig. 8.7). They have an angle or curve in a plane perpendicular to that of the handle. They are used mainly for 'lateral cutting'. Lateral cutting instruments are made 'left' and 'right', by having the curve/angle, either on the right or left. Holding the instrument with its blade down and away, the instrument having curve of the blade directed to the right is a 'right' instrument and similarly for the 'left'.



Fig. 8.7: Double plane instrument

Single plane instruments are considered safer to use; however, double-ended instruments are more efficient, and also less instrument exchange is required. Blades of different widths can also be placed on double-ended instruments.

iii. ***Contra-angling:*** In many instruments, shank has bends at one or more points to angle the blade relative to the long axis of the handle. The extent of this arrangement depends on the length of the blade and the degree of angulation in the shank. Accordingly, the working point is moved out of line with the axis of the handle. If the working point is moved out of line with the axis of handle, the instrument can be out of balance in lateral cutting motion.

To avoid this, instruments are designed to have one or more angles in the shank, placing the working point within 3.0 mm from the axis of the handle. This principle of design is called contra-angling.

The length of the blade is determined by the depth of the cavity and the blade angle is determined by the accessibility requirements. Greater angles are required posteriorly and in proximal cavities of anterior teeth. Contra-angling also provides better access of the operating field.

iv. ***Single beveled instruments:*** Instruments which have a bevel only on one side of the instrument blade are known as single beveled instruments. If the instrument is beveled on the side away from the shaft, it is called 'distally beveled'. If beveled on the side of the blade towards the shaft, it is called 'mesially beveled'.

If the bevel is on the right side of the instrument blade, it is a right-sided instrument and if the bevel is on the left side, it is a left-sided instrument.

When these types of instruments have no angle in the shank or an angle of 12.5 centigrade or less, they are used in push (direct cutting) and scraping motions (chisel). If this angle in the shank exceeds 12.5 centigrade, the instruments could be used in pull (distally beveled) and push (mesially beveled) motions (hoe) (Fig. 8.8A to C).

v. ***Bibeveled instruments:*** Instruments beveled on both the sides of the blade, which meet together to form the cutting edge are called bibeveled instruments, e.g. ordinary hatchet (Fig. 8.13C).

vi. ***Triple-beveled instruments:*** Beveling the blade laterally, along with the end, forms three

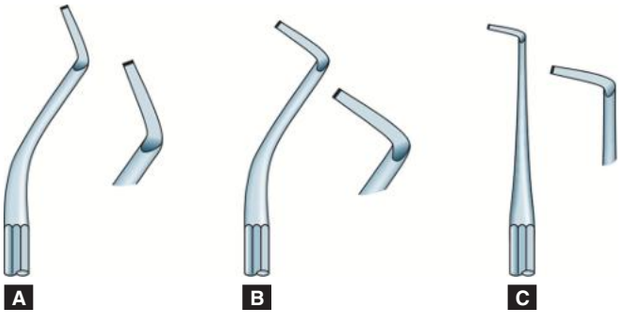


Fig. 8.8: Transition from chisel **(A)** to hoe (**B** and **C**); by increasing blade angulation

distinct cutting edges, called triple-beveled instrument, e.g. angle former (Fig. 8.17C).

vii. ***Circumferentially beveled instruments:*** These are double-planed instruments, where the blade is beveled at all peripheries, e.g. spoon excavator (Fig. 8.12B).

viii. ***Single-ended and double-ended instruments:*** Single-ended instruments are confined to only one specific function; while double-ended instrument incorporates the right and left or mesial and distal forms of the instrument in the same handle.

The individual hand cutting instruments are described as follows.

a. *Chisels*

The chisels, designed for planing and cleaving, are characterized by a blade that terminates in a cutting edge formed by a one-sided bevel. Cutting edge of a chisel is usually at right angle to the shaft.

The types of chisels are:

i. ***Straight chisel:*** These have a straight blade in line with the handle and shank. The cutting edge is on one side only (Fig. 8.9A).

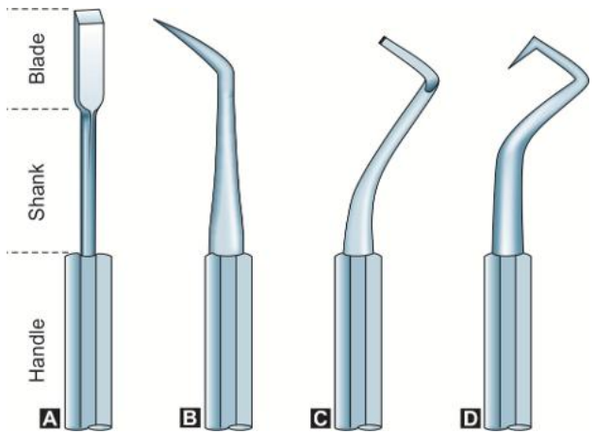


Fig. 8.9: Chisels: **(A)** Straight chisel; **(B)** Monoangle chisel (blade at an angle to shaft; mesial/distal); **(C)** Binangle chisel (two angles between shaft and blade); **(D)** Triple angle chisel (three angles in the shank)

ii. ***Monoangle chisels:*** The blade is placed at an angle to the shaft; may be mesially (standard) or distally (reverse) beveled (Fig. 8.9B).

iii. ***Binangle chisels:*** These have two angles between the shaft and the blade; may be mesially or distally beveled (Fig. 8.9C and 8.10).

iv. ***Triple-angle chisels:*** These have three angles in the shank. These are usually used to flatten pulpal floor may be mesially or distally beveled (Fig. 8.9D).

b. *Excavators*

Excavators are used for excavation of caries and sharpening/refinement of the internal walls of the cavity preparation.

The types of excavators are:

i. ***Hoe:*** A hoe is a form of chisel in which the angle of the blade nearly approaches a right angle, i.e. 25 centigrade. It is a single-plane instrument, which can be distally or mesially beveled and used with pull motion (Fig. 8.8).

Hoe is used for cutting mesial and distal walls of premolars and molars.

ii. ***Spoon:*** The cutting edge is ground to a semi-circular circumferential bevel and sharpened to a thin edge. These are available in pairs; the blade of one curved to the right, and the blade of other curved to the left. The spoon excavators are double-planed instruments with right or left cutting movement (Fig. 8.11).

Spoon is used for removal of decayed dentin.

iii. ***Cleoid-Discoid:*** It is similar to the spoon excavator, except that the blade resembles a claw; hence, the name 'cleoid'. It is used in carving amalgam and excavating decay from areas of difficult access (Fig. 8.12A, B).

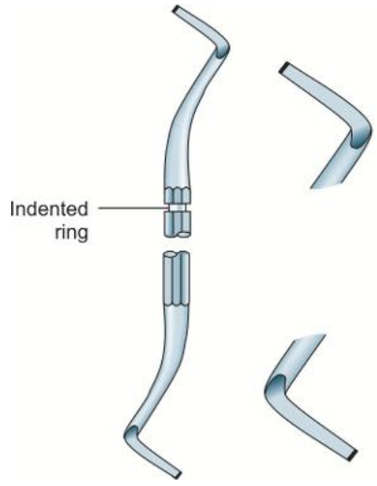


Fig. 8.10: Binangle chisel

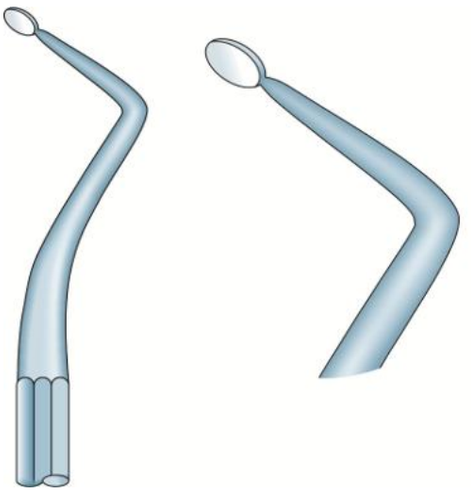


Fig. 8.11: Spoon excavator



Fig. 8.12: Cleoid-Discoid instrument

c. *Modified Forms of Chisels*

The modifications are designed to perform specific functions. The types of modified chisel are:

i. ***Hatchet:*** Hatchet is used for splitting/cleaving undermined enamel in proximal cavities and on buccal and lingual walls, where it is not possible to use a chisel. The smaller sizes are used in anterior teeth and larger sizes are used in posterior teeth (Fig. 8.13A to C).

ii. ***Gingival margin trimmer (GMT):*** It is a modified form of hatchet (Fig. 8.14A, B). The modifications (Fig. 8.15A to F) of the basic hatchet design are:

- Cutting edge of a hatchet is perpendicular to the axis of the blade while cutting edge of a gingival margin trimmer is at an angle other than a perpendicular angle (GMT) to the axis of the blade (so a 4-unit formula).
- Hatchet has a straight blade; the blade of a GMT is curved (Fig. 8.16A, B).

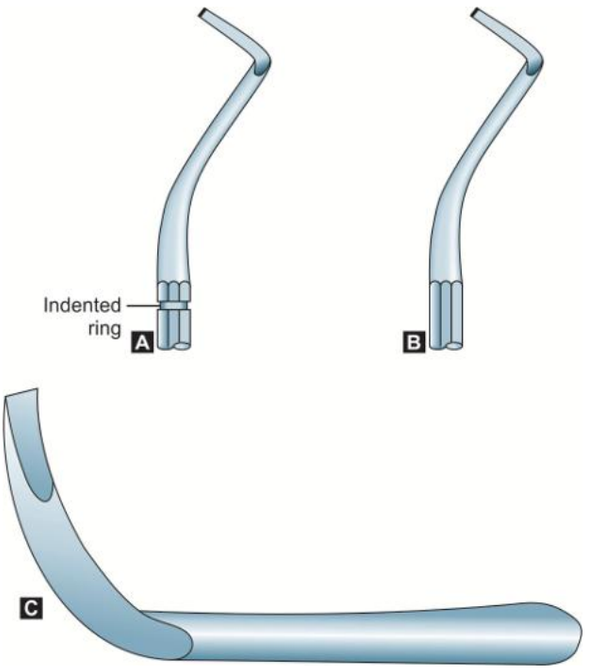


Fig. 8.13: **(A and B)** Hatchet (cutting edge perpendicular to the axis of blade); **(C)** Bevel on two sides (bibeveled)



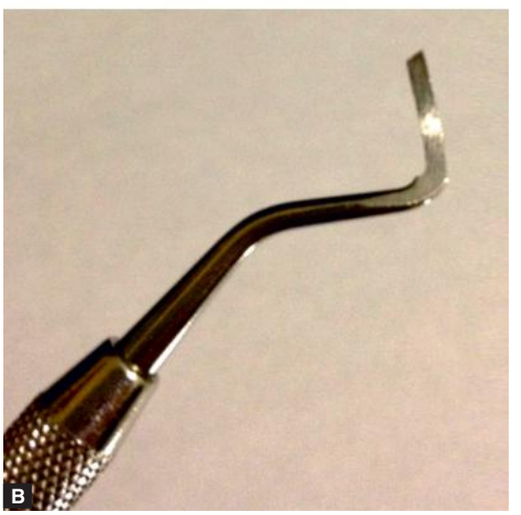


Fig. 8.14: Gingival margin trimmer

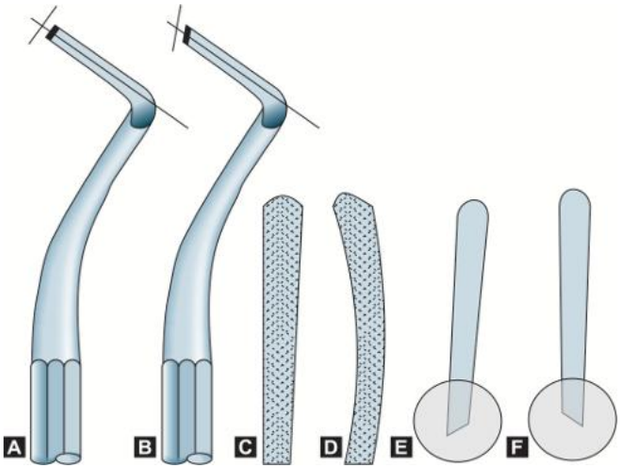


Fig. 8.15: Hatchet **(A)** Compared with gingival margin trimmer **(B)**; **(C)** Straight blade; **(D)** Curved blade; **(E)** Right cutting; **(F)** Left cutting

- Hatchet is a single plane instrument, while GMT is a double plane instrument (primarily a lateral cutting instrument). It is paired with right- and left-sided bevels.
- There are two pairs of these instruments, consisting a set of four. In a given size, each pair has a right and left beveled instruments. The cutting edges of one pair make an acute angle with that edge of the blade furthest from the handle. These are distal gingival margin trimmer. A typical distal GMT is the 10-95-6-12, the second number in instrument formula may vary from 95 to 100. The cutting edge of other makes an acute angle with that edge of the blade nearer to the handle. These are mesial gingival margin trimmer. A typical mesial GMT is the 10-80-12, the second number in instrument formula may vary from 75 to 85.

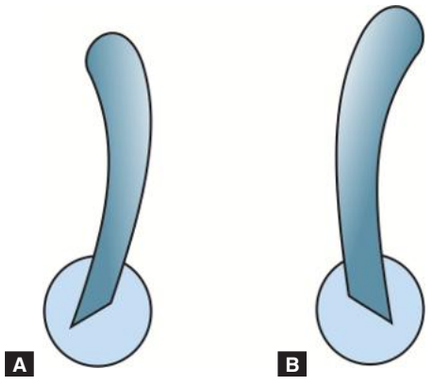


Fig. 8.16: Gingival margin trimmer (curved cutting edges): **(A)** Right; **(B)** Left

GMT is used for creating a bevel at the gingival cavosurface margin and to bevel axiopulpal line angle in Class II cavity. Distal GMT is used for distal surface and the mesial GMT is used for mesial surface. It is used for trimming the margins of walls of cavity preparation.

It is also used for sharpening internal walls of the cavity preparations.

iii. ***Angle former:*** It is a modified form of chisel. In this instrument, the primary cutting edge is sharpened at an angle to the axis of the blade. The angle of cutting edge to the blade axis is usually 80–85° (4–unit formula). Blade of the angle former is beveled on the sides to form three cutting edges. The acute cutting angle being directed to the right or left makes the angle former a paired instrument. Right of the pair is identified by an indented ring. It is a single plane instrument (Fig. 8.17A, B).

Angle former is used to accentuate line and point angles in the internal outline form. It is also used in cavity preparation for cohesive gold to establish retention form.

iv. ***Wedelstaedt chisel:*** It is like a straight chisel, but with a slight vertical curvature in its shank. It is beveled on one side only, which can be placed mesially or distally. It is a single plane instrument with three cutting motions viz. vertical, right and left. It is used for cleaving undermined enamel and for shaping walls. The mesially beveled can be used in push movements and the distally beveled can be used in pull motions (Fig. 8.18).

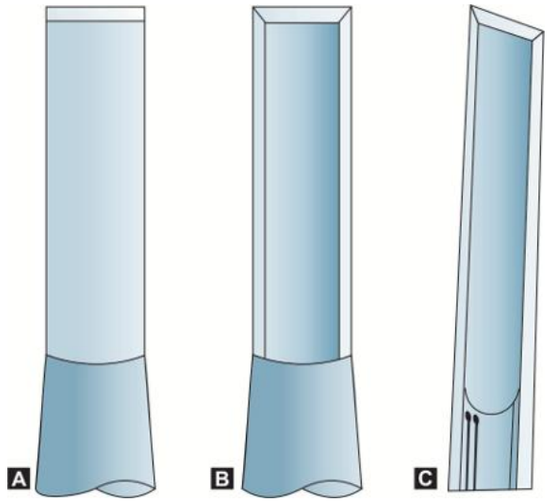


Fig. 8.17: Transition from chisel **(A)** to angle former **(B)** by creating cutting edge on the lateral sides; **(C)** Showing bevel on three sides (triple bevel)



Fig. 8.18: Wedelstaedt chisel

d. *Miscellaneous*

i. ***Knives:*** Nibs of these instruments carry knife-edged faces on one side only. The knives are made in various sizes and shapes (Fig. 8.19).
They are used for trimming off excess filling material on the gingival, facial or lingual margins of a proximal restoration or trimming and contouring the cavosurface margins of a Class V restoration.
The knives used for specific purposes are:
- *Wilson's knife:* The nib is at perpendicular angle to that of the shaft, so that it can be introduced interproximally for use.
- *Stein's knife:* It has a trapezoidal nib and is used in direct gold restorations for contouring and margination.

ii. ***Dental files:*** Nib in files can be foot-shaped, hatchet- shaped or parallelogram-shaped with serrations (Fig. 8.20). If the serrations are directed away from the handle, it is a 'push' file and if the serrations are directed towards the handle, it is a 'pull' file. The routinely used hand-cutting instruments are shown in Fig. 8.21.



Fig. 8.19: Dental knife

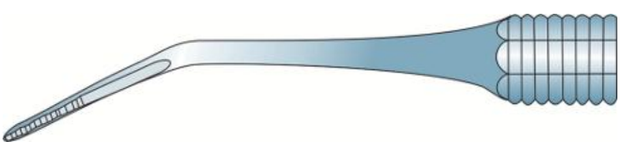


Fig. 8.20: Dental file

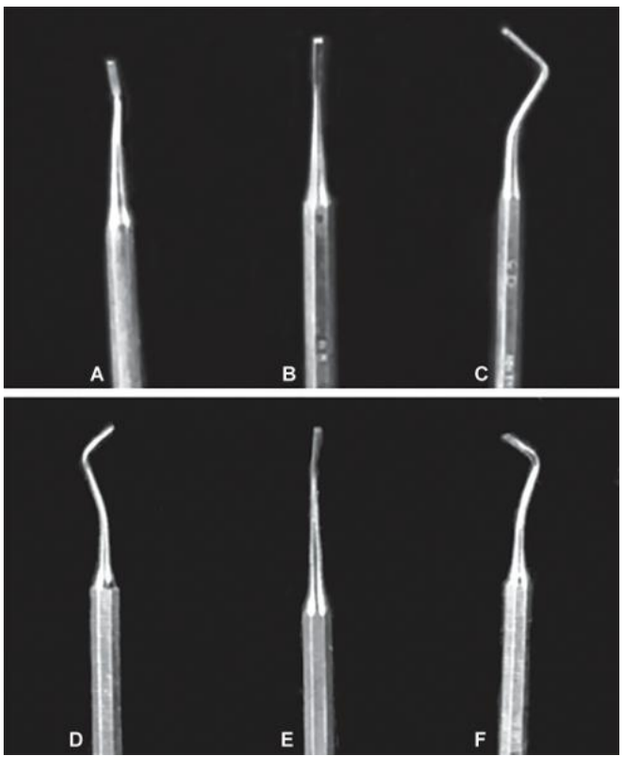


Fig. 8.21: Instruments: **(A)** Binangled chisel; **(B)** Straight chisel; **(C)** Hatchet; **(D)** Left gingival margin trimmer; **(E)** Hoe; **(F)** Right gingival margin trimmer

Advantages of hand-cutting instruments

- Self-limited in cutting enamel; do not cut sound enamel, only enamel undermined by loss of dentin.
- Remove large pieces of undermined enamel quickly; saving time and effort.
- No vibration or heat during cutting; painless and no adverse effects on the tooth tissues.
- Most efficient means of precise cutting, especially when cutting is needed adjacent to important anatomy.
- Create smooth surface.
- Long lifespan; can be re-sharpened.

B. NON-CUTTING (GENERAL) INSTRUMENTS

a. Diagnostic Instruments

i. ***Mouth mirror*:** Mouth mirror is usually a single-ended instrument, made of steel; disposable plastic handle mirrors are also available. The mirror is detachable and can be fixed in handle (socket type end). Various types and sizes are available to suit operator needs [one surface, two surface and concave (magnifying) mirrors] (Fig. 8.22A to C).

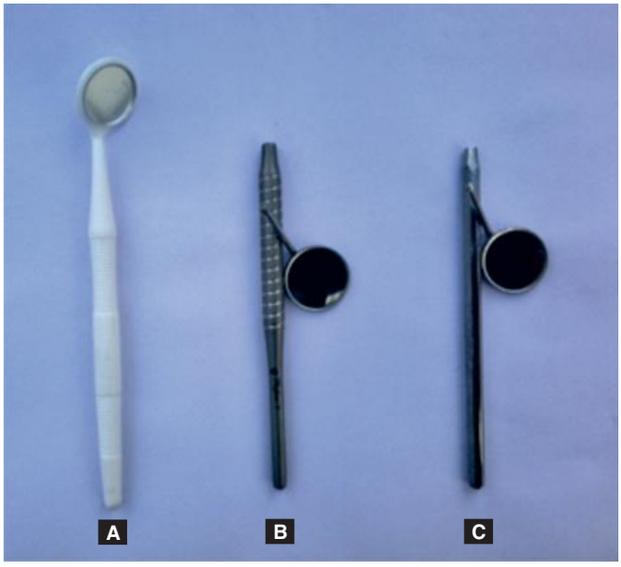


Fig. 8.22: Mouth mirrors: **(A)** Disposable mirror; **(B)** Conventional mirror; **(C)** Mirror with lens

Functions

- Provide indirect light at the target tissues
- Provide indirect vision
- Retract tongue/cheeks during tooth preparation and other procedures
- Facilitates intra-oral photography

ii. *Probes:* Probes are both single-ended and double-ended. These are available in variety of shapes and sizes; the shape of the pointed tip can be straight, curved, right angled, double curved etc. (Fig. 8.23). A few probes are calibrated, with millimetre markings (e.g. periodontal probe). These are specially designed to measure periodontal depth. Other such varieties are Naber's probe, CPITN probe, etc. (Fig. 8.24A to C).

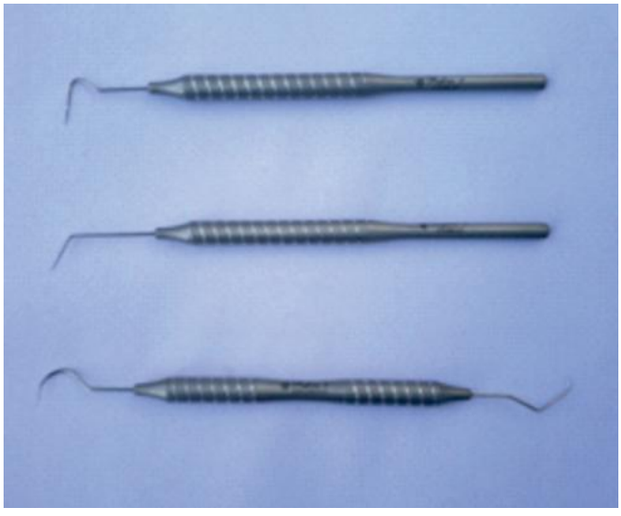


Fig. 8.23: Probes (different shapes)

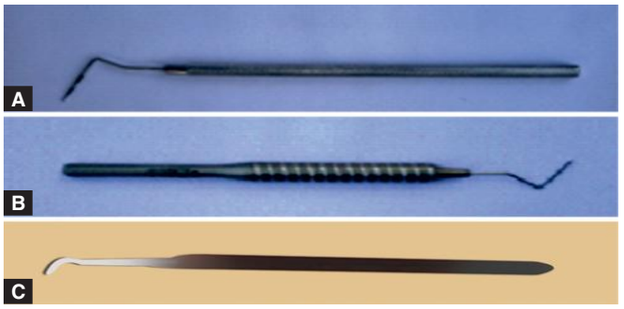


Fig. 8.24: Probes: **(A)** Periodontal probe; **(B)** CPITN probe; **(C)** Naber's probe

Functions

- Examination of caries, calculus, etc.
- Examination of irregularities on tooth surfaces
- Can be used to puncture an intraoral abscess
- Useful in pin-pointing the furcation involvement

iii. *Tweezers:* Tweezers are available in different sizes, with angulated working end (Fig. 8.25). They usually have plain tip; however, serrated tips are also available. Tweezers may have lock, which facilitates holding of cotton, gauze, etc.

Functions

- Used to grasp cotton or gauze
- For carrying small burs and other items

b. Restorative Instruments

i. *Condensers:* Condensers are used to condense the filling material into the prepared cavity. These are of different shapes and sizes, depending upon the area of the tip. The tip can be smooth or serrated and may be round or parallelogram. Mainly

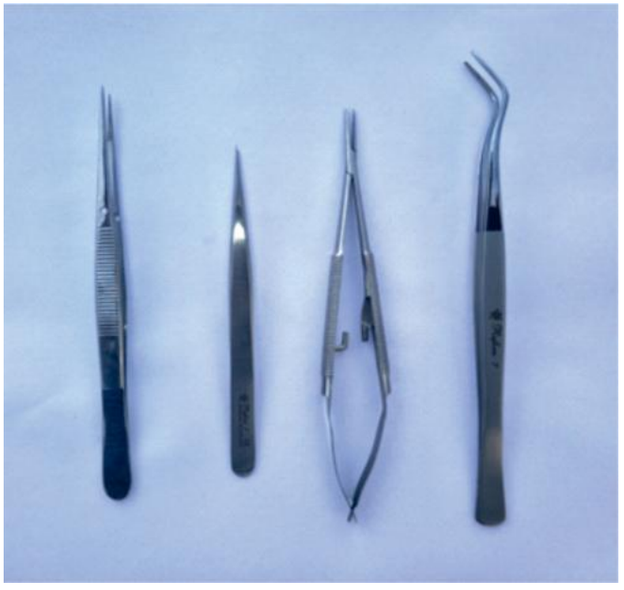


Fig. 8.25: Tweezers (different shapes)

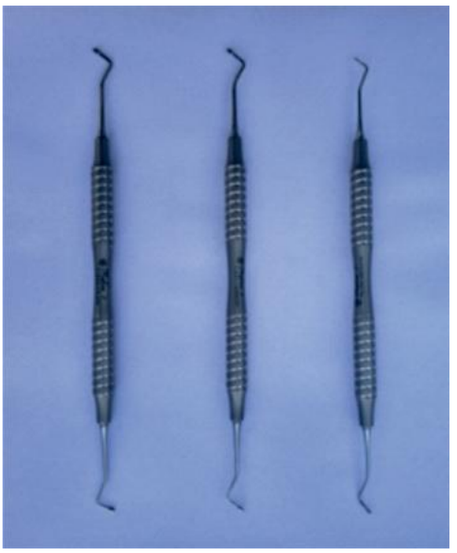


Fig. 8.26: Condensers

used for condensing amalgam and also for smoothening base in the cavity (Fig. 8.26)

ii. *Carvers:* Carvers are used to shape the restorative materials (plastic materials). Carvers are mainly used to create anatomy of tooth surface in amalgam and composites. Various types of carvers are available in different sizes (Fig. 8.27).

iii. *Burnishers:* Burnishers are used to smoothen the restoration to remove minor irregularities after compaction of the material. In gold restorations, burnishers are used to merge the gold onto the bevels, so as to achieve gap-free margins with the cavity walls. Burnishers are available in different sizes; mostly they are single-ended instrument (Fig. 8.28).

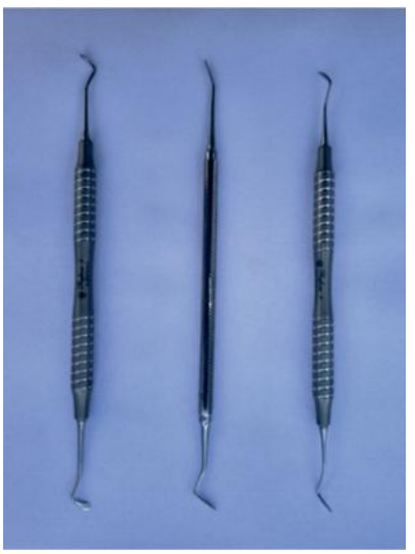


Fig. 8.27: Carvers

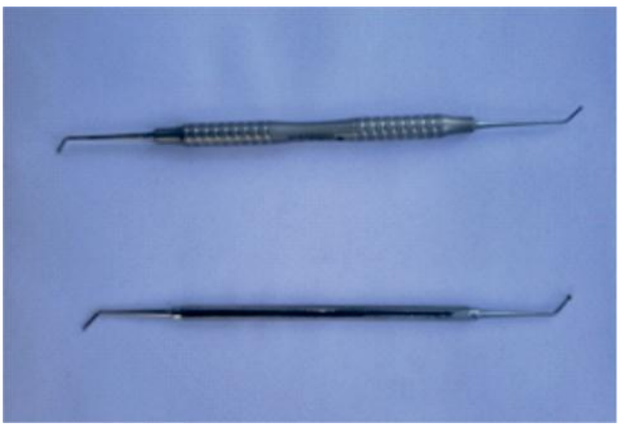


Fig. 8.28: Burnishers



Fig. 8.29: Plastic instruments

iv. *Plastic instrument:* Plastic instruments are used to carry restorative material, when it is in plastic stage, to the inner cavity preparation. Plastic instruments are available in different sizes and shapes (Fig. 8.29).

c. Finishing/Polishing Instruments

A variety of strips, made of plastic and steel, are used to finish proximal area of the restoration; categorized in non-cutting instruments (used with hand affecting minor abrasive action). Finishing strips are available in different sizes to suit the proximal area of the restoration (Fig. 8.30A, B).

A hand instrument kit is shown in Fig. 8.31.

INSTRUMENT GRASPS

Proper holding of instrument improves clinical efficiency and decreases physical stress.

The main grasps used with hand instruments are as follows:

i. *Pen grasp:* It is similar to holding a pen, except that the pulps of the thumb, 1st and 2nd fingers

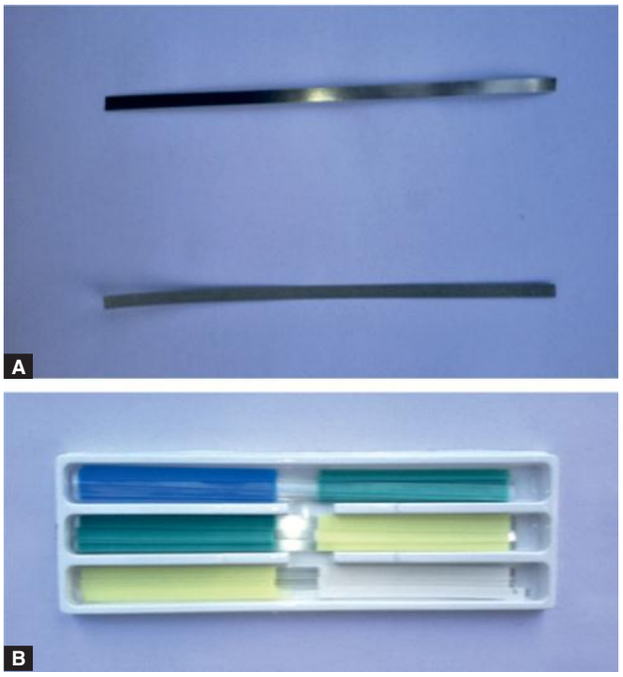


Fig. 8.30: Finishing strips: **(A)** Metal; **(B)** Plastic

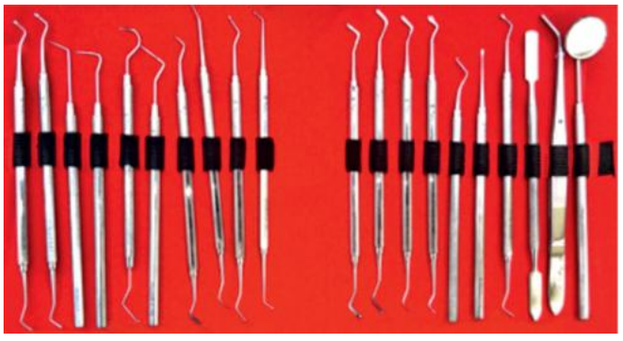


Fig. 8.31: Hand instrument kit

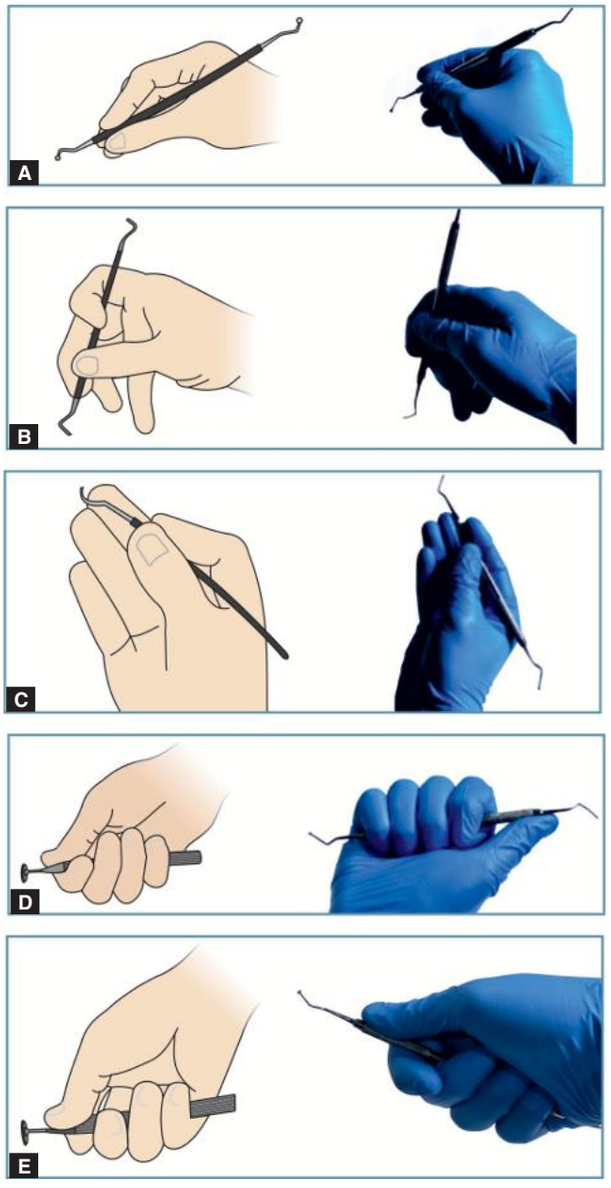


Fig. 8.32: Instrument grasps: **(A)** Pen grasp; **(B)** Modified pen grasp; **(C)** Inverted pen grasp; **(D)** Palm grasp; **(E)** Modified palm grasp

contact the instrument, while the tip of the 3rd and 4th fingers are placed on the adjoining teeth (as rests).

The position of 2nd finger is important for good control and thrust to the instrument; due to greater length of the 2nd finger, the application point for the force will be near the working point of the instrument (Fig. 8.32A, B). The pen grasp and the modified form are used universally for most clinical procedures.

ii. *Inverted pen grasp:* This is similar to the pen grasp, but the hand is rotated so that the palm is facing upwards. It is usually used in upper teeth. It is not used frequently (Fig. 8.32C).

iii. *Palm and thumb grasp:* It is similar to holding a knife. The handle is placed in the palm of the hand and grasped by the four fingers, while the thumb rests on an area other than that being operated on. A supporting rest provided by the thumb is necessary because digital control is somewhat insufficient (Fig. 8.32 D).

This grasp is used when the thumb rests at some point distant from the operating site. It may be useful on maxillary teeth, particularly the right side, when working from the right rear chair position.

iv. *Modified palm and thumb grasp:* When it is feasible to rest the thumb on the same tooth being

operated or on a tooth immediately adjacent, the modified palm and thumb grasp may be used (Fig. 8.32E).

The handle of the instrument is held between pulps of the thumb and the 1st and 2nd fingers. The 3rd and 4th fingers are about half closed; they contact the handle under the 1st joint of each finger and press the handle against the distal area of the palm. It provides greater freedom of movement.

Rests

Rests are used to steady the hand during operating procedures and these act as guards when thrust force is applied.

The rests should be on the arch being operated on and preferably on the same quadrant. The closer the rest to the operating area, the more reliable the effects are. In some instances, it is impossible to secure a rest on tooth structure. In such cases, soft tissues can be used; however, soft tissues do not provide the reliable rest (Fig. 8.33A to F).

Guards

Guards are finger positions opposite the one using the instrument, to steady the area being operated on. The guards provide protection in case the instrument accidentally slips off the working surface (Fig. 8.33A to F).

These should be used particularly when adjacent teeth are not available as rests or when rests are obtained on soft tissues.

SHARPENING OF HAND INSTRUMENTS

With frequent use and sterilization, the cutting edges lose sharpness and become dull.

Use of instruments with dull cutting edges causes pain, prolongs operating time and reduces preciseness in cavity preparation. Therefore, it is essential that all cutting instruments be sharp.

Sharpening is carried out by reducing the bulk of the metal at the cutting edge, following the original configuration of the bevel.

There are many types of sharpening instruments, viz. Arkansas stones, mechanical sharpener and diamond stones (used in the handpiece).

The principles followed for sharpening are:

- Do not sharpen dirty instruments
- Establish the proper bevel angle (usually 45–60°) and the desired angle of the cutting edge to the blade before placing the instrument against the stone and maintain these angles while sharpening.
- Use a light stroke or pressure to minimize frictional heat.
- Use a rest or guide, whenever possible.
- Remove as little metal from the blade as possible.
- The unbeveled side of the blade should be finished to remove the fine edges that may be created.
- Keep the sharpening stones clean and free of metal cuttings.

The sharpening can be manual on stationary stone or by mechanical means.

i. ***Stationary stone technique:*** The size of the Arkansas stone should be at least two inches wide

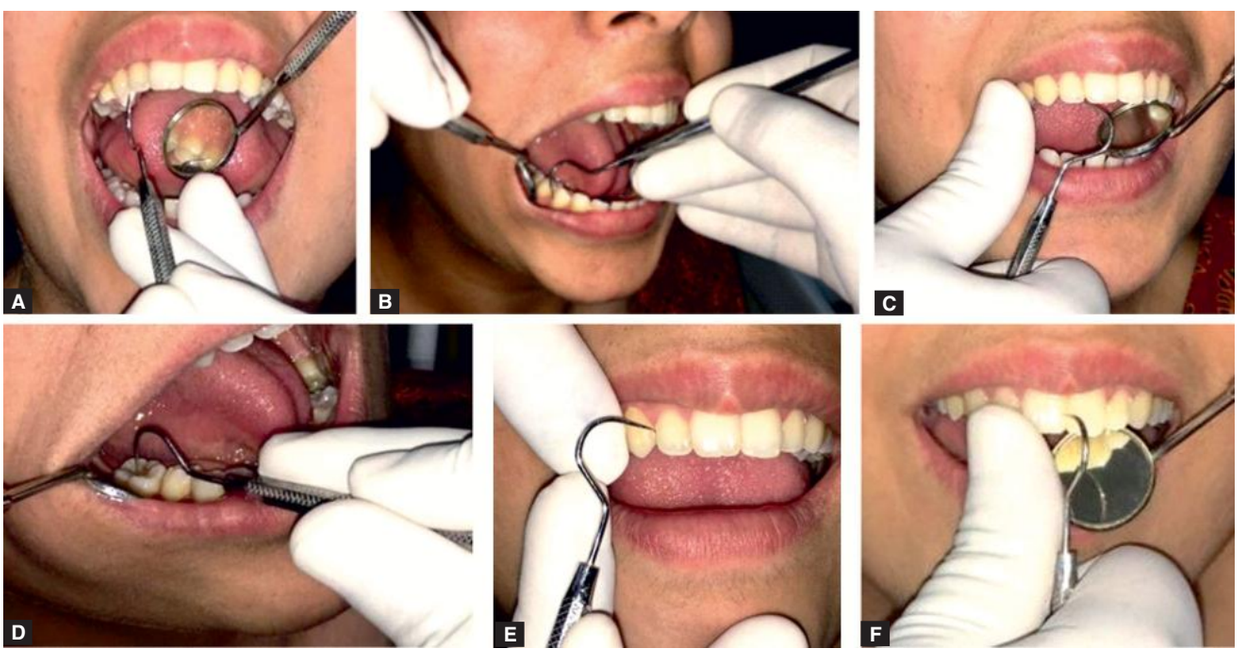


Fig. 8.33: Various forms of rests and guards

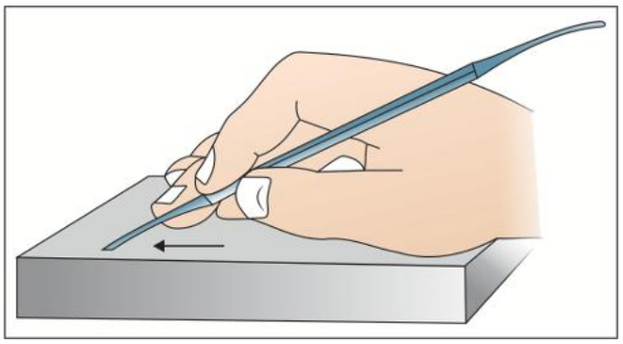


Fig. 8.34: Sharpening of instruments (moving instrument in line with the edge (arrow)



Fig. 8.35: Gear-driven handpiece (contra-angle and straight)

and five inches long. Before using, a thin film of light oil should be applied on the working surface. Stone should be laid on a flat surface and should not tilt while sharpening. Instrument should be grasped with a pen grip, to avoid its rotation while sharpening. To ensure stability, third and fourth fingers are used as rest and guide along a flat surface to prevent rolling/dipping of the instrument (Fig. 8.34).

ii. *Mechanical techniques:* While using mechanical sharpener, the proper angle of the cutting edge to the blade should be established before activating the motor.

Handpiece stones are used for instruments with curved blades, especially for inside curve of the blades. The handpiece should run at a slow speed and instrument should be held lightly with a pen grasp. The third and fourth fingers should be used as rests.

ROTARY-DRIVEN INSTRUMENTS

The term 'rotary instruments' refers to a group of instruments that turn on an axis to perform a work, such as cutting, abrading, burnishing, finishing or polishing tooth tissues or a restoration.

The development and evolution of rotary driven instruments is a continuous process. There are three types of rotary instruments, i.e. handpiece, dental burs and abrasive instruments.

A. DENTAL HANDPIECES

a. Gear-Driven Handpiece

The three conventional designs of handpieces are straight, contra-angle and prophylaxis angle. Rotary cutting instruments are inserted into a chuck at the front of the handpiece (Fig. 8.35).

Conventional handpieces were initially designed to operate at speeds under 5000 rpm. By the use of speed increasing transmissions, it is possible to obtain speeds of 100,000 rpm. The gear-driven angle has an automatic lubricating system.

Gear-driven handpieces are versatile, being capable of a wide speed range and use. All operative procedures can be accomplished; however, they function best at low speed because of so many moving parts with metal to metal contact. Proper maintenance is of utmost importance to prevent excessive heat, vibration and wear within the handpiece.

b. Water-Driven Handpiece

A hydraulic-driven turbine handpiece called 'turbojet', which can operate at speeds up to 100,000 rpm. The turbojet is designed as a compact unit, which runs on electricity. The sound-proof cabinet contains a motor, water pump, water reservoir and the necessary plumbing for circulating the water. The small inner tube carries water under high pressure to rotate a turbine in the handpiece head and the larger outer tube returns the water to the reservoir. It is recirculated time and again.

c. Belt-Driven Handpieces

A belt-driven handpiece, called 'Page-Chayes' can operate at speeds above 100,000 rpm. All gears were eliminated by having a small belt, which run inside the handpiece over ball bearing pulleys.

Improved models of the belt-driven design are the Page-Chayes 909 and the Twin 909.

Belt-driven handpieces are relatively free of maintenance problems because the bearings have factory sealed lubrication.

d. Air-Driven Handpieces

Air-driven handpiece with speed of approximately 300,000 rpm consists of a handpiece, control box, foot control and the connector hoses (Fig. 8.36A to C). When the foot control is activated, compressed air is carried by a flexible hose to the handpiece. The air is



Fig. 8.36A: Air-driven handpiece (high speed)



Fig. 8.36B: Air-driven ultra-speed



Fig. 8.36C: Ultra-speed fiberoptic

blown against the blades of a small turbine to produce rotation while the greater part is exhausted at the back of the handpiece. As the lateral workload is increased, the speed decreases. Air turbines have low torque and will stall at lateral workloads. This is an excellent safety feature since excessive pressure cannot be applied to rotary instrument.

e. Mini-Head/Micro-Head Handpiece

Mini-head/micro-head handpiece is used in pediatric procedures and in smaller openings of mouth (Fig. 8.37). Handpiece at an angle is used during surgical procedures (Fig. 8.38).



Fig. 8.37: Mini- and micro-head handpiece



Fig. 8.38: Surgical handpiece (angulated head)

Speed Ranges and Uses

Low/conventional speed : Below 6000 rpm
High/intermediate speed : 6000–100,000 rpm
Ultra/super speeds : Above 100,000 rpm

i. ***Low speed:*** Low speed is used for excavating caries with round burs, refining cavity preparations with sand paper discs, marginating gold restorations and for polishing procedures.

ii. ***High speed:*** High speed can be used for cavity preparations. Many procedures, such as placement of retentive grooves and bevels are best performed at high speeds. This speed range is preferred where more positive sense of touch is needed as a guide.

iii. ***Ultra-speed:*** Ultra-speed is desirable for bulk reduction, obtaining outline form and removing metal restorations. Cavity preparations can be initiated at ultra-speeds, but final finishing may require high speed only.

Common Features

Various types of rotary cutting instruments have certain design features, which are common, viz., shank, neck, and head (Fig. 8.39A, B).

i. ***Shank:*** The shank is that part of the rotary instrument that fits into the handpiece, accepts the rotary movement from the handpiece and controls the alignment and concentricity of the instrument. The shanks can be:
- Straight
- Latch type
- Friction grip

Straight shank is a simple cylinder, held in the handpiece by a metal chuck. Because of its long

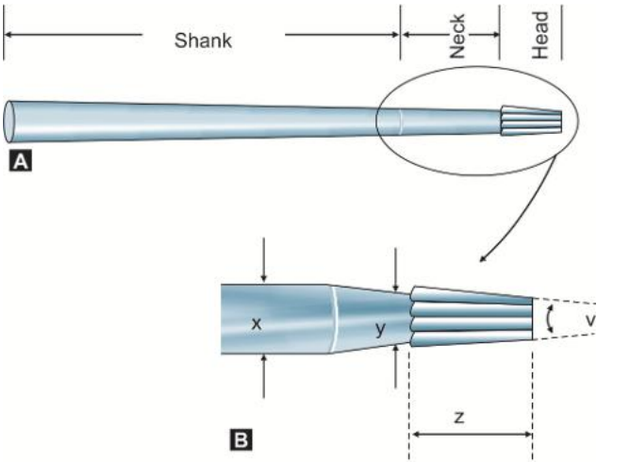


Fig. 8.39: **(A)** Parts of a rotary cutting instrument; **(B)** Enlarged view shows (x) shank diameter, (y) neck diameter, (z) head length, and (v) taper angle

length, its access to posterior region of the mouth is difficult. It is mostly used extraorally for finishing and polishing laboratory restorations.

The latch type instruments fit into the metal tube within the handpiece. These are retained by a latch that slides into the groove placed at the shank end. Their short length allows easy access and is primarily used at low and medium speeds for finishing procedures.

Friction grip shank is a simple cylinder, held in the handpiece by friction between the shank and a plastic or metal chuck. Dimensional accuracy of the shanks is important in these instruments. The overall length is smaller than the latch type instruments, thus improving access. These are designed to be used at high speeds.

ii. ***Neck:*** The neck is the intermediate part of the instrument that connects the shank to the head. It tapers from the shank diameter up to the head. The neck dimensions may be a compromise between the need for a large cross-section to provide strength and a small cross-section to improve access and visibility. The neck transmits rotational and translational forces to the head.

iii. ***Head:*** The head is the working part of the instrument whose cutting edges perform the desired functions. The head shows great variations in design; can be bladed or abrasive. The size and shape may also vary.

Neck tapers from the shank diameter up to the head. The main function of neck is to transmit rotational and translational forces to the head.

B. DENTAL BURS

Bur is defined as "*a rotary cutting instrument with cutting heads of various shapes and two or more sharp-edged blades, used as a rotary grinder*". It is used for various purposes, such as, finishing restorations, surgical removal of bone and tooth preparation.

Metallurgy

Metallurgically, dental burs are of two types: Stainless steel burs and tungsten carbide burs. Steel burs are cut from steel blank parallel to the long axis of the bur. Steel burs perform well at low speeds, and dull rapidly at high speeds or when cutting enamel.

Tungsten carbide burs are obtained by alloying in which complete fusion of the constituents does not occur. The tungsten carbide powder is mixed with powdered cobalt/nickel under vacuum. A blank is then formed and the bur is cut from it with a diamond tool. Carbide burs perform better than steel burs at all speeds, especially at high speeds. Carbide is harder than steel and, therefore, does not dull rapidly. However, carbide is also more brittle and more susceptible to fracture.

In most of the modern burs, a combination of the two is used. The carbide head is attached to a steel shank and neck by welding. A major drawback with the use of these burs is that steel necks bend, in comparison to carbide necks, which fracture under sudden stress.

Even slight bending of bur, produces increased vibrations because of increased *run out*. A 'dull' bur produces more heat with reduced cutting efficiency.

Classification

There are various systems for the classification of burs.

1. According to mode of attachment to the handpiece—latch type or friction grip type.
2. According to composition—stainless steel burs, tungsten carbide burs or a combination.
3. According to motion—right or left bur. A right bur is one which cuts, when it revolves clockwise and a left bur is one which cuts when revolving anticlockwise.
4. According to length of head—long, short or regular.
5. According to use—cutting burs or finishing burs.
6. According to shape—round, inverted cone, pear-shaped, wheel-shaped, tapering fissure, straight fissure, end cutting, etc.

Bur Shapes

The term 'bur shape' refers to the contour or silhouette of the bur head. The basic head shapes are the round, inverted cone, pear, straight fissure, tapering fissure and end-cutting (Fig. 8.40A to F).

a. ***Round bur:*** A bur with a spherical head is a round bur. It is used for initial access preparation, removal of caries, and for the placement of retentive grooves.

b. ***Inverted cone bur:*** A bur with a rapidly tapering cone head (small end of the cone is directed towards the shank). It is used for establishing

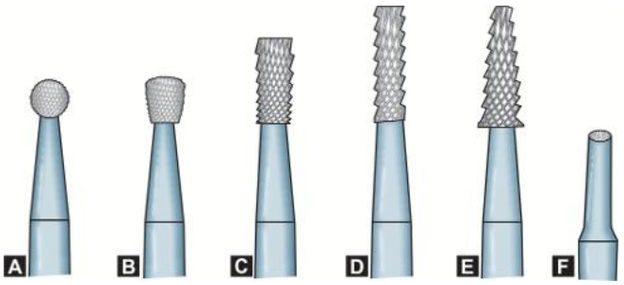


Fig. 8.40: Different shapes of commonly used dental burs: **(A)** Round; **(B)** Inverted; **(C)** Pear; **(D)** Straight fissure; **(E)** Tapering fissure; **(F)** End cutting

undercuts in cavity preparations and providing indirect bevels.

c. ***Pear-shaped bur:*** A bur with a slightly tapering cone (small end of the cone directed towards the shank). It is used in Class I cavity preparations for gold foil. A long length pear bur is advocated for cavity preparations for amalgam.

d. ***Straight fissure bur:*** A bur with the head shape of an elongated cylinder. It is used for amalgam cavity preparations.

e. ***Tapering fissure bur:*** A bur with a tapering cone head (small end of the cone directed away from the shank). It is used for inlay and crown preparations.

f. ***End-cutting bur:*** This bur is cylindrical in shape, with just the end carrying the blades. It is used for carrying the preparation apically without axial reduction.

Bur Sizes

The original numbering system (SS White) grouped burs by 9 shapes and 11 sizes. The ½ and ¼ designations were added later (Table 8.2). Crosscut burs were indicated by adding 500 to the equivalent non-crosscut size (e.g., no. 56 bur with crosscuts was designated as 556). Similarly, an end-cutting bur was indicated by adding 900 to the equivalent size.

Over the years, the numbering is modified (Table 8.3).

Modifications

The major modifications include:

- Reduced use of crosscuts
- Extended heads on fissure burs
- Rounding of the sharp tip angles

Crosscuts are notches present on the blade of the instrument to obtain adequate cutting effectiveness at low speeds (Fig. 8.41). At higher speeds, they tend to produce unduly rough surfaces, so many of the crosscut burs designed to be used at low speeds have been replaced by equivalent non-crosscuts burs to be used at high speeds.

With the use of high speeds, carbide fissure burs have been introduced that have extended head lengths (two to three times the normal length).

Table 8.2: Original bur head sizes (adapted from SS White Dental Manufacturing Company)

Head shapes	*Head diameter in mm*												
	0.5	0.6	0.8	1.0	1.2	1.4	1.6	1.9	2.1	2.3	2.5	2.8	3.0
Round	1/4	1/2	1	2	3	4	5	6	7	8	9	10	11
Wheel		11½	12	13	14	15	16	17	18	19	20	21	22
Cone		22½	23	24	25	26	27	28	29	30	31	32	33
Inverted cone		33½	34	35	36	37	38	39	40	41	42	43	44
Bud		44½	45	46	47	48	49	50	51	52	53	54	55
Straight fissure (flat end)		55½	56	57	58	59	60	61	62				
Straight fissure (pointed end)		66½	67	68	69	70	71	72	73				
Pear		77½	78	79	80	81	82	83	84	85	86	87	88
Oval		88½	89	90	91	92	93	94	95				

Table 8.3: Standard bur head size (carbide & steel) (adapted from SS White Dental Manufacturing Company)

Head shapes	*Head diameter in mm*													
	0.5	0.6	0.8	1.0	1.2	1.4	1.6	1.9	2.1	2.3	2.5	2.8	3.0	3.3
Round	1/4	1/2	1	2	3	4	5	6	7	8	9	10	11	
Wheel		11½	12		14		16							
Inverted cone		33½	34	35	36	37	38	39	40					
Plain fissure		55½	56	57	58	59	60	61	62					
Round crosscut				502	503	505	506							
Straight fissure crosscut			556	557	558	559	560	561	562	563				
Tapered fissure crosscut				700	701	702	703							
End cutting fissure				957	958	959								
Round finishing				A	B	C	D		200	201			202	203
Oval finishing									218	219			220	221
Pear finishing									230		231	232		
Flame finishing				242	243	244	245	246						



Fig. 8.41: Cross-cut bur, lateral view: (×) spiral angle

Sharp angles in the bur head can result in areas of increased stress concentration in the teeth and increase their tendency to fracture. Rounding of the sharp angles enhances the strength of the tooth and also the instrument lasts longer.

The most commonly used burs and their dimensions are listed in Table 8.4.

Design of a Dental Bur

The actual cutting action of a bur takes place at the edge of the blade present on the bur head. The bur head consists of uniformly spaced blades with depressed areas in between them (Fig. 8.42). These depressed areas are called as the *flute* or the *chip* spaces. The number of blades on a bur is usually even because even numbered blades are easy to produce

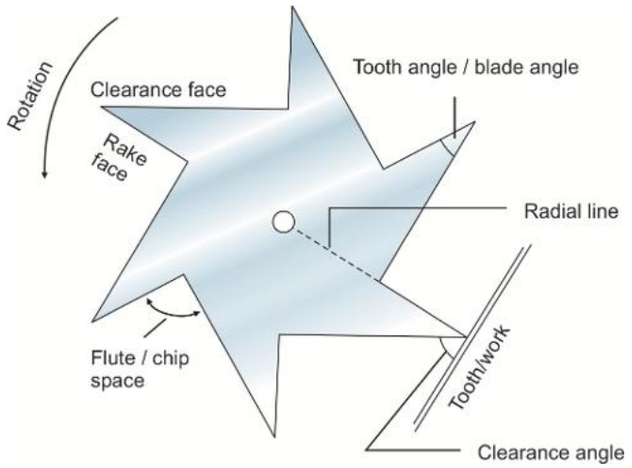


Fig. 8.42: Cross-section of a bur

in the manufacturing process. The blades on a cutting bur are usually 6 to 8 to 10 and those on a finishing bur are usually 12 to 40. The greater the number of blades, smoother is the cutting action at low speeds. At high speeds, no more than one blade cuts effectively at any one time.

i. ***Blade:*** The projection on the bur head is known as the blade or the tooth. This terminates in the cutting edge. It has two surfaces: The blade face/rake face and the blade back/flank/clearance face. Rake face is the surface of bur blade on the leading edge and clearance face is the surface of the bur blade on the trailing edge.

ii. ***Blade angle/tooth angle:*** It is the angle between rake face and the clearance face. In case a land is present, it is the angle between rake face and the land.

iii. ***Rake angle:*** The rake angle is the most important design characteristic of a bur blade. It is the angle between the rake face and the radial line (line

Table 8.4: Key dimensions and names of commonly used burs

Manufacturer's number	*ADA size number*	*Shape*	*Head diameter (mm)*	*Head length (mm)*
¼	¼	Round	0.50	0.40
½	1/2	Round	0.60	0.48
2	2	Round	1.00	0.80
4	4	Round	1.40	1.10
33S	–	Inverted cone	0.60	0.45
33.5	33.5	Inverted cone	0.60	0.45
169	169	Tapered fissure	0.90	4.3
169	169L	Elongated tapered fissure	0.90	5.6
329	329	Pear normal length	0.70	0.85
330	330	Pear normal length	0.80	1.00
245 =	330L	Pear elongated length	0.80	3.00
271 =	171	Tapered fissure	1.20	4.0
272 =	172	Tapered fissure	1.60	5.0

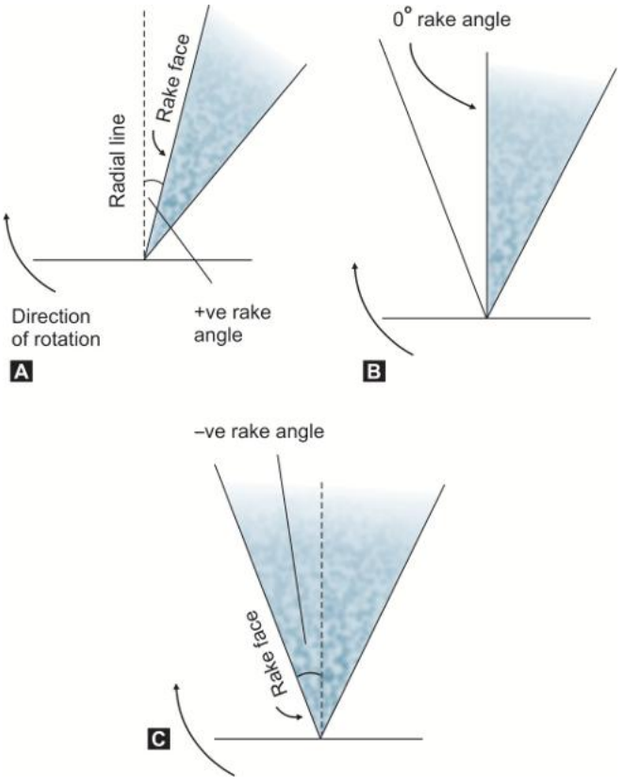


Fig. 8.43: Rake angle: **(A)** Positive rake angle; **(B)** Radial rake angle; **(C)** Negative rake angle

connecting the centre of the bur and the blade). Accordingly, it can be a positive, negative or a zero rake angle (Fig. 8.43A to C). A rake angle is negative when the rake face is ahead of the radial line. It is considered as positive when the rake face trails the radial line. A zero rake angle also known as the 'radial rake angle' is seen when the rake face and the radial line coincide with each other.

iv. *Land:* The plane surface immediately following the cutting edge is called the land (Fig. 8.44).

v. *Clearance angle:* The angle between the clearance face and the work (e.g. tooth surface) is called the clearance angle. If a land is present on the bur, the clearance angle is divided into two: Primary clearance angle, i.e. the angle between land and the work and a secondary clearance angle, i.e. the angle between clearance face and the work. If the clearance face is curved, it is known as radial clearance (Fig. 8.45A, B).

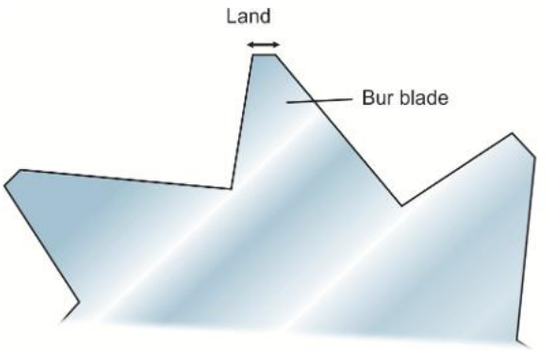


Fig. 8.44: Bur blade with a land

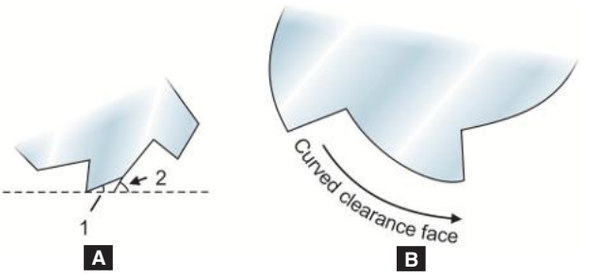


Fig. 8.45: **(A)** A land on the bur blade divides the clearance angle into two: (1) primary clearance angle, (2) secondary clearance angle; **(B)** Bur with a curved clearance face (radial clearance)

Cutting Efficiency of a Bur

The cutting efficiency of a bur depends upon following factors, viz:

1. *Rake angle, clearance angle and blade angle:* The positive rake angle is directly proportional to the cutting efficiency of the bur (more positive angle, better cutting). However, a positive rake angle has two major drawbacks: (i) It reduces the bulk of the bur blade, as a result, the edge of blade can easily curve, flatten or even fracture during use; (ii) A positive rake angle produces a larger chip and tends to clog the flute space.
   The clearance angle eliminates friction between the cutting edge and the work and prevents the bur from digging excessively into the tooth structure. An increase in the clearance angle, however, reduces the blade angle, thereby decreasing the bulk of the bur blade.
   Increasing the blade angle reinforces the cutting edge and reduces the chances of the blade edge to fracture.
   Carbide burs have blades with slightly negative rake angle and a blade angle of approximately 90°. Their clearance faces are usually curved to provide a small clearance angle near the cutting edge and a greater clearance space.
2. *Neck diameter:* An instrument with small neck diameter will be unable to resist lateral forces. A large neck may interfere with the visibility and restrict access. As the head of the bur increases in length and diameter, the movement exerted by the lateral forces also increases and the neck needs to be larger.
3. *Spiral angle and crosscuts:* Burs with small spiral angles are preferred at high speeds as small angles produce more efficient cutting (Fig. 8.41).

Crosscuts are notches in the blade edges to increase the cutting effectiveness at low and medium speeds. The greater the length of a blade, greater is the force required to initiate cutting. Crosscuts tend to reduce the total length of the bur blade that is cutting at any one time. This increases the force per unit area and thereby reduces the pressure required to initiate cutting.

As each crosscut blade cuts, it leaves ridges of tooth structure in place of the notches. Because the notches in the two succeeding blades are not in line, ridges left by one blade are removed by the other. However, at high speeds, only one blade cuts effectively, leaving behind the ridges and hence, a rough surface.

4. ***Concentricity and run-out:*** Concentricity is a direct measurement of the symmetry of the bur head. It measures how closely a single circle can be passed through the tips of all the blades. It is an indication as to whether one blade is longer or shorter than the others.

   Run-out is a dynamic measurement of the maximum displacement of the bur head from its axis of rotation while the bur runs. The average clinically acceptable run-out is 0.023 mm. Run-out occurs, if the head is not in line with the axis of the bur, if the bur neck is bent and if the bur is not held straight in the handpiece. Run-out tends to increase the vibrations of the bur and also removes excessive amount of tooth structure. Tooth removal with run-out is inefficient, inaccurate and heat generating. Run-out is more important clinically as compared to concentricity.

The primary cause of vibrations during cutting is runout; an important feature clinically. Runout determines the dimensions of hole vis-a-vis the diameter of the bur head. (Burs usually cut holes larger than the head diameter.)

5. ***Heat treatment:*** Heat treatment is used to harden a bur made of soft steel. This process preserves the cutting edge, subsequently increasing the life of the bur. Carbide burs do not require heat treatment.
6. ***Influence of load:*** The load is the force exerted by the operator on the tool head and not the pressure induced in the tooth during cutting. Load should be in the range of 1000–1500 gm (2–3 pounds) for low speeds and from 60–120 gm (2–4 ounces) for high speeds. The cutting effectiveness of the bur is reduced when the load applied is below the minimum required and also is reduced when the load applied is above the maximum required.
7. ***Influence of speed:*** At a given load, the rate of cutting increases with the rotational speed but not in direct proportion. The rate of cutting is more at speeds above 30,000 rpm; however, it has been found that at speeds greater than 1,50,000 rpm, the rate of cutting is nearly the same. There is also a minimum rotational speed for a given load below which the tool will not cut.
8. ***Number of teeth or blades:*** The number of teeth in a dental bur are usually limited to 6–8. Since the load is distributed among the blades, decreasing the number of blades increases the force on one blade and also increases the size of the chip removed. This tends to reduce the clogging tendency since the flute space becomes larger. However, a major drawback with burs having less teeth is that the tendency for bur tooth wear is more and the cutting life is reduced. Also, a fewer number of bur teeth increases the tendency for vibration.

   A fissure bur with straight flutes also produces less temperature rise than one with spiral flutes. This may be due to the formation of large chips with the use of a straight fluted bur. Large chips carry away more heat as compared to smaller chips.

Polymer Burs

A bur material, which removes only the infected dentin and not the affected dentin was introduced by SS White (USA) as polymer bur (Smart Prep). The bur has a unique flute design, constructed from polyether-ether-ketone (PEEK) with particular hardness and wear resistance (Fig. 8.46). This enables the bur to remove only soft caries-infected dentin, leaving the caries-affected dentin intact. The bur usually works in slow speed (500–800 rpm) and quickly dulls when encounters the highly calcified caries-affected dentin.

Polymer bur

- Main manufacturer: SmartPrep (SS White, USA)
- Made of a softer polyamide/imide polymer material
- Used with slow speed (slow handpiece)
- Harder than carious dentin and softer than healthy dentin
- Remove infected dentin (caries) only
- Preferably used for one time
- Less effective than carbide bur in caries removal



Fig. 8.46: Polymer burs



Fig. 8.47: CVD coated diamond bur

The polymer burs are single use, self-limiting instrument, which wears away rather than cutting sound dentin. The operator can feel the resistance of the healthy dentin. The polymer burs elicit no pain, therefore, can be effectively used without anesthesia.

Another quality, CVD-coated diamond bur has been introduced for ultraconservative cavity preparation (Fig. 8.47).

C. ABRASIVE INSTRUMENTS

The head of abrasive instruments consists of small particles of a hard substance held in a matrix of softer material. This matrix that holds the particles together is called binder. Different materials used as binder are ceramic, metal, rubber, shellac, etc. Rubber and shellac wear away easily and are used for delicate abrasion, like finishing and polishing.

Cutting occurs at a large number of points, where the hard particles protrude from the binder. As the instrument is used, the particles got worn off and new particles replace the older ones, thereby providing a continuous supply of particles. The abrasive should be so distributed that the surface of the tool wears away evenly. The wide spacing between the particles prevents the packing or clogging of the resultant debris.

Abrasive instruments can be grouped into diamond abrasives and other abrasives. Diamond instruments have great clinical impact in operative dentistry because of their long life and great effectiveness in cutting enamel and dentin. Other abrasive instruments include use of silicon carbide, boron carbide, aluminum oxide, garnet and sand for producing grinding wheels, discs or stones, etc.

a. Diamond Abrasive Instruments

Diamond abrasive instruments are preferred over tungsten carbide burs because of their greater resistance to abrasion, lower heat generation and longer life. The instrument consists of three parts: A metal blank, powdered diamond abrasive and a binding material (Fig. 8.48).

Like other rotary instruments, it has three parts—head, neck and shank. The shape and size of the metal blank control the shape and size of the instrument.

The abrasive diamonds may be natural or synthetic that are applied in multiple layers by electro-deposition, sintering or microbrazing.

The binding agent serves the purpose of holding the abrasive particles together as well as attaches the particles to the metal blank. Ceramics are most commonly used binding agents.

Classification

The diamond abrasive instruments are classified on the basis of average particle sizes of the abrasive, viz:

- Coarse grit diamond burs (125–150 μ particle size)
- Medium grit diamond burs (88–125 μ particle size)
- Fine grit diamond burs (60–74 μ particle size)
- Very fine grit diamond burs (38–44 μ particle size)

Head Shapes and Sizes

Diamond instruments are available in a variety of shapes and sizes that mostly correspond to the burs (except very small numbers, as ¼, ½ round burs). Because of abrasive coating over the shank, diamond instruments cannot be of very small sizes as compared to burs. However, additional number of diamond sizes are available (Fig. 8.49).

Due to the lack of any standard nomenclature for diamond instruments; these are selected visually by their shape.

Abrasive Efficiency

The clinical performance of diamond abrasive instruments depends upon a number of factors, such as:

1. ***Size of the abrasive particles:*** The larger the size of the particles, deeper is the penetration on the surface of the work; hence, rapid removal of

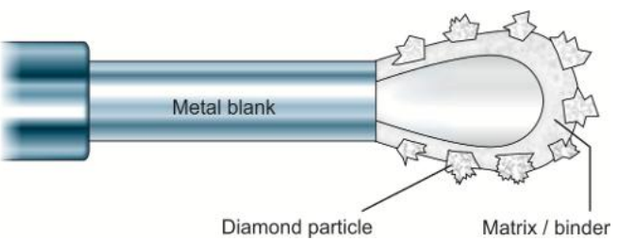


Fig. 8.48: Schematic diagram of dental diamond instrument

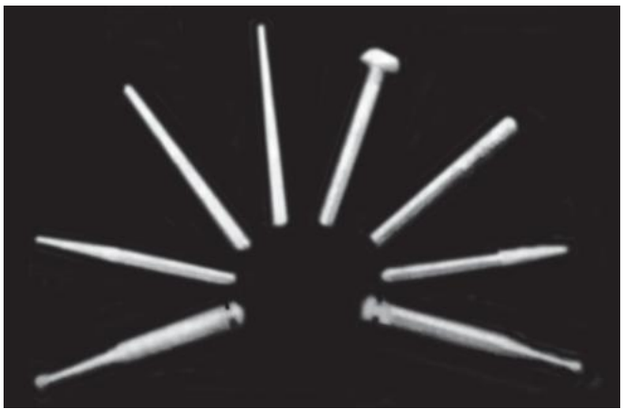


Fig. 8.49: Diamond instruments of varying shapes

material is achieved with coarse grit points as compared to medium or fine grit points. However, a few authors found no difference in the cutting efficiency of coarse and medium grit diamond points.

2. ***Shape of the abrasive particles:*** The abrasive particles should be irregular in shape for greater efficiency. Irregular particles present a sharp edge and better cutting; whereas, smooth particles present a flat face, hence, less cutting.
3. ***Density of the abrasive particles:*** Density of the particles refers to the number of abrasive particles per unit area. In high density, the particles are closely packed, whereas in low density, the particles are widely spaced. Coarse grit diamond points have a low density as compared to fine grit burs.
   The low density points may have less grinding rate because of the deeper penetration of their diamonds into enamel causing early particle dislodgement.
4. ***Hardness of the abrasive particles:*** The hardness of the abrasive particle should be greater than the hardness of the work; only then the abrasive indent the surface and remove the material.
5. ***Clogging of the abrasive surface:*** Clogging of the spaces between the abrasive particles affects grinding, because this partially blocks the penetration of the abrasive particles into the surface. Clogging is enhanced when the particles are close together. Use of coolants during grinding may wash away the debris and prevent clogging. The diamond points should be cleaned after every use.
6. ***Speed and pressure:*** Proper speed and pressure are the major factors determining the life of a diamond point. The usual cause of failure of abrasive instruments is when excessive pressure is applied to increase their cutting rate at inadequate speeds. This results in the loss of diamonds, decreasing their cutting efficiency.
7. ***Miscellaneous:*** These include uncontrolled operating parameters, like differences in dental hard tissues, rotation speed, turbine air pressure, differences in handpieces, etc.

Disposable Diamond Abrasives

The centers for disease control and prevention recommend that all dental instruments that penetrate soft or hard tissues or come into contact with the oral tissues must be sterilized after each use to prevent cross-contamination (Fig. 8.50). The "single patient use" or disposable diamond bur is introduced that minimizes the risk of cross-contamination of blood-borne pathogens. Some of the disposable diamond burs are marketed as Cobra, Monosteryl, Neo, Patriot, SS White and Spring.

b. Other Abrasive Instruments

Many types of abrasive instruments other than the diamonds are used in dentistry. They are used for shaping, finishing and polishing restorations both in the clinic and in the laboratory. The cutting surfaces of the head are composed of abrasive particles held in a continuous matrix of softer material. They can be of two type—molded abrasives and the coated abrasives.

i. *Molded abrasive instruments* are manufactured by molding or pressing a uniform mixture of abrasive and matrix around the roughened end of the shank (Fig. 8.51A). The abrasive is distributed throughout the matrix such that new particles are exposed by continual wear. Rigid molded instruments have rigid polymer (grinding and sharpening procedures); whereas, soft molded instruments use flexible matrix materials (used for finishing and polishing procedures).



Fig. 8.50: Solution used for disinfection of burs

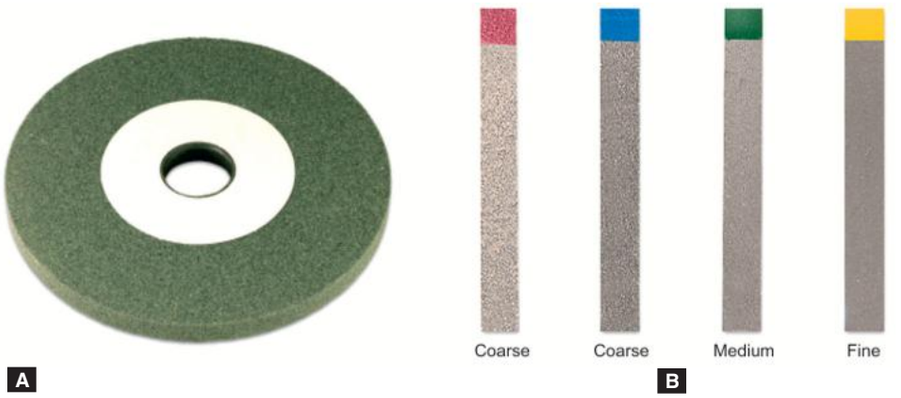


Fig. 8.51: Abrasive instruments: **(A)** Molded; **(B)** Coated

Table 8.5: Evolution of rotary cutting instruments

Year	*Instrument*	*Speed (rpm)*
1800	Hand-rotated instruments	300
1871	Foot engine	700
1880	Electric engine	1,000
1915	Dental unit	5,000
1942	Diamond cutting instruments	5000
1948	Tungsten carbide burs	12,000
1950	Ball-bearing handpieces	25,000
1955	Water-turbine handpiece	50,000
1955	Belt-driven handpiece (Page-Chayes)	150,000
1957	Air-turbine handpiece	250,000
1994	Air-turbine handpiece (improved)	300,000
1998	Fiberoptic air-turbine handpiece	>3,00,000

rpm = revolutions per minute

ii. *Coated abrasive instruments* are mostly discs which have a thin layer of abrasive cemented to a flexible backing (Fig. 8.51B). Unlike the molded instruments, coated abrasives have to be discarded, once they wear off. They are used mainly for finishing of restorations, especially the proximal restorations.

Evolution of rotary cutting instruments is summarized in Table 8.5.

MECHANISM OF CUTTING

Cutting can be measured in terms of effectiveness and efficiency. Cutting effectiveness is the rate of removal of tooth structure in mm/min or mg/sec. Effectiveness does not include side effects, such as heat or noise. Cutting efficiency is the percentage of energy, which produces cutting. Cutting efficiency is decreased when some amount of energy is wasted as heat or noise. Increase in rotational speed increases both effectiveness and efficiency; but also has disadvantages, like heat, noise and vibration.

Rotary instruments can cut by two mechanisms—blade cutting and abrasive cutting.

Blade cutting refers to the use of a bladed instrument or cutting by any instrument in a blade-like fashion. For the blade to cut, it must be sharp, have higher hardness and modulus of elasticity than the material being cut. The chips created by cutting accumulate in the clearance space between the flutes, until they are washed away or removed by centrifugal force.

A lot of heat is produced with the use of these bladed burs. Distortion of the tooth structure produces heat. Frictional heat is also produced by the chips rubbing against blade tip along the cut surface of the tooth.

Abrasive cutting/grinding utilizes bonded or coated abrasive instruments for removing small particles of the substrate. Grinding instruments contain randomly arranged abrasive particles that are sharp and have negative rake angles.

When these instruments are used on the substrates, some part may be removed as chips and some part may flow laterally around the cutting tip forming ridges. This type of cutting is less efficient; hence, burs are preferred for cutting ductile materials, such as

dentin. Abrasive instruments are most efficient when used to cut brittle materials such as enamel.

Advantages of Rotary Cutting

- Cutting is relatively easy and accessible.
- Quick and time saving.
- Easy to control the cutting
- Provide tactile perception of the extent of cutting
- Debris can be removed by washing.
- Familiar and well known

Disadvantages of Rotary Cutting

- Rotary cutting may cause pain.
- Vibration may cracks or fracture tooth structure.
- Patients not comfortable with the noise produced.
- Constant use and sterilization can deteriorate the cutting effect.
- Produce lot of heat (potential pulp damage).
- May lead to overcutting, if the operator loses control.
- Difficult to achieve precise cutting.

PATIENT AND OPERATOR POSITIONS

The operator should maintain proper position and posture during treatment, because:

i. It has a definitive bearing on the correct performance.
ii. It affects the comfort and co-operation of the patient.
iii. It influences the health of the operator; the operator is less likely to get strain, fatigue, be more efficient and less chances of getting musculo-skeletal disorders.

Position of Dental Chair

The adjustment of dental chair is an important feature. If the chair is incorrectly placed, the accuracy, and precision necessary to perform are lessened. The patient is also subjected to unnecessary, un-comfortable position.

The important reasons for placing the patient correctly in the dental chair are:

i. *Comfort of the patient:* When the patient is sitting comfortably and relax, the confidence in the operator grows; extremely desirable for the operator.
ii. *Proper position for operating:* The operation can be completed more accurately, swiftly and easily when access to the field of operation is good.

Modern dental chairs are designed to provide total body support and comfort in any position. These chairs rely on raising, lowering and letting the backrest for accommodating patient's position. The letting of backrest only causes sliding of the patient in the chair, which does not happen when both the backrest and seat are moved together in the programmable chairs.

While seating the patient in the chair, the important features are:

- Before the patient is brought into the operatory, the moveable appliances should be moved out so that the patient can sit without twisting, turning, bending, stooping, etc.
- The height should be low, the backrest upright and the armrest adjusted to allow the patient to get into the chair. After patient is seated, the armrest is positioned to its normal position.
- The adjustable headrest cushion and the chair back is positioned to support the head and elevate the chin slightly. In this position, neck muscle strain is at a minimum and swallowing is easiest. The chair is then adjusted to place the patient in recline position.
- The patient should be lowered to position that places the treatment site as close as to the operator's elbow level as possible.
- The chair design should provide access and maximum working is to the operator.
- It should be placed at convenient position with adjustable control switches.
- The chairs equipped with programmable operating positions are preferred. To improve infection control, chair with foot switches for patient positioning is recommended.

Position of Patient

The common patient positions for restorative dental procedures are:

i. Upright position
ii. Almost supine position
iii. Recline position.

However, the choice of patient position varies with the operator, the type of procedure and the area of oral cavity involved in the operation.

i. In upright position, the backrest is at 90° to the floor (mainly used for initial diagnosis) (Fig. 8.52).
ii. In almost supine position, the patient's head, knees and feet are approximately at same level. Patient head should not be lower than the feet except in case of when the patient is in syncope (Fig. 8.53).
iii. In recline position, the chair is reclined at 45°. Consequently, mandibular occlusal surfaces are almost at 45° to the floor (Fig. 8.54).

As a rule, when operating in the maxillary arch, the maxillary occlusal surfaces should be oriented approximately perpendicular to the floor; the mandibular arch and the mandibular occlusal surfaces should be almost at 45° to the floor.



Fig. 8.52: Supine position



Fig. 8.53: Almost supine position



Fig. 8.54: Recline position

After the operation is completed, the chair should be brought back to the upright position and height should be lowered, so that the patient can leave the dental chair easily, gracefully, preventing undue strain or loss of balance.

Operating Positions

The operating positions depend upon location of the operator in relation to the patient position. For right-handed operator, there are essentially three positions—right front, right and right rear; similarly reverse should be applied for the left-handed operators. A fourth position, 'direct rear' has been used for certain areas of the oral cavity. For understanding, positions of operator can be related to the clock (an imaginary circle is drawn keeping the patient head at the center of the circle). The numbering to circle is given similar to clock with the top of the circle at 12 o'clock. Accordingly, the operator's positions (right-handed operator) can be 7 o'clock (right front), 9 o'clock (right), 11 o'clock (right rear) and 12 o'clock (direct rear). For left-handed operator, it can be 5 o'clock (left front), 3 o'clock (left) and 1 o'clock (left rear).

i. ***Right front position:*** This position facilitates working on (i) mandibular anterior teeth, (ii) mandibular posterior teeth on right side and (iii) maxillary anterior teeth. The pen grasp is the preferred grasp for this position. To increase ease and visibility, the patient's head may be turned towards the operator.

ii. ***Right position:*** The operator sits exactly right to the patient (Fig. 8.55). This position is convenient for working on (i) facial surfaces of maxillary right posterior teeth, (ii) facial surfaces of mandibular right posterior teeth and (iii) the occlusal surfaces of the mandibular right posterior teeth. The pen grasp is used from this side.

iii. ***Right rear position:*** This is the preferred position for the operator (Fig. 8.56). Most areas of the oral cavity are accessible from this position either using direct or indirect vision. The left arm is positioned around the patient's head when working from this position. The areas included are: Palatal and incisal (occlusal) surfaces of maxillary teeth, and mandibular teeth especially on the left side. Direct vision may be used on these teeth, however, the use of mirror will improve the visibility with the reflection of light and retraction. Some stools have backrest with curved extensions that offer additional body support.



Fig. 8.55: Right position (9 o'clock)



Fig. 8.56: Right rear position (11 o'clock)



Fig. 8.57: Direct rear position (12 o'clock)

iv. *Direct rear:* This position has limited application (Fig. 8.57). Here the operator sits directly behind the patient and looks down over the head of the patient. This position is used for operating on lingual surfaces of mandibular anterior teeth. In the maxillary arch, mainly mirror is used for the preparation.

Operating Stools

Various varieties of stools are available for the operator and the assistant (Fig. 8.58A, B).

The features required for operating stools are:

- Should be on casters for mobility
- Should be sturdy and well balanced
- Well padded with smooth cushion edges and adjustable up and down
- The backrest should be adjustable forward, backward as well as up and down along with full back support. The assistant's stool should have a foot ring for footrest. Operator stools may not have a footrest.



Fig. 8.58: Operating stools

Position of Assistant

- The selected position for the assistant is essentially the same as for the operator, except that the stool is 4 to 6 inches high for maximal visual access.
- Assistant should sit as close as possible to back of the patient's chair. The thighs should be parallel and feet firmly placed on foot support ring at the base of the stool. These positions can be adjusted according to specific needs.
- The instrument tray should be so placed, as to allow easy access to the instruments and materials.

Implications of Operating Positions

Dental practice may produce physical and mental strain on the operator. Every effort should be made to minimize the strain and to keep the body at the highest state of efficiency through adopting correct posture, eliminating all deteriorating forces, which can reduce nerve power and physical endurance. Usually sitting position is preferred in modern dentistry to relieve stress on operator's legs and also support the back.

The correct position of operator for working in the oral cavity is as follows:

- Operator should sit back on the cushion, using the entire seat and not just the front edge. The upper body should be positioned so that the spinal column is straight or bent slightly forward and supported by the backrest.
- The back and chest are held in upright position with the shoulders squared to promote circulation and proper breathing. Position that creates unnecessary curvature of the spine or slumping of the shoulders should be avoided.
- The thighs should be parallel to the floor and lower legs perpendicular to the floor. Feet should be flat

on the floor. If the seat is high, circulation to the legs will be compromised.

- Operator should breathe slowly and naturally through nose, avoiding inhaling the patient's breath or exhaling into patient's face.
- The bodily contact with the patient should be avoided. The operator should not rest forearms on the patient's shoulders and hands on the face of the patient.
- Proper working distance, similar to that for book reading should be maintained during operative procedures. This will lead to increased co-operation and confidence.
- When the patient is properly positioned, the distance between eyes of the operator and working field should be 12"–14", however, inaccessible tooth preparations may require closer proximity for adequate visibility.
- Operator should not use patient's chest as instrument trolly.
- Patient's head can be rotated backward, forward or from side-to-side for operator's ease, access and visibility.
- Operator should leave left hand free for retraction using mouth mirrors.
- When operating for an extended period, change in position of operator for a short time reduces muscle strain and lessen fatigue.

Area of Instrument Exchange

For right-handed operator, there are four zones of working area, i.e. (i) operator's zone (lies between 7 o'clock to 12 o'clock), (ii) assistant's zone (between 2 to 4 o'clock), (iii) static zone (between 12 to 2 o'clock) and (iv) transfer/exchange zone (between 4 to 7 o'clock). The transfer of instrument between the operator and assistant should occur in the exchange zone below the patient's chin and several inches above the patient's chest. Instrument should not be exchanged over the patient's face.

All instruments and materials are located in the assistant's zones. Static zone is a non-traffic area where other equipment can be placed.

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